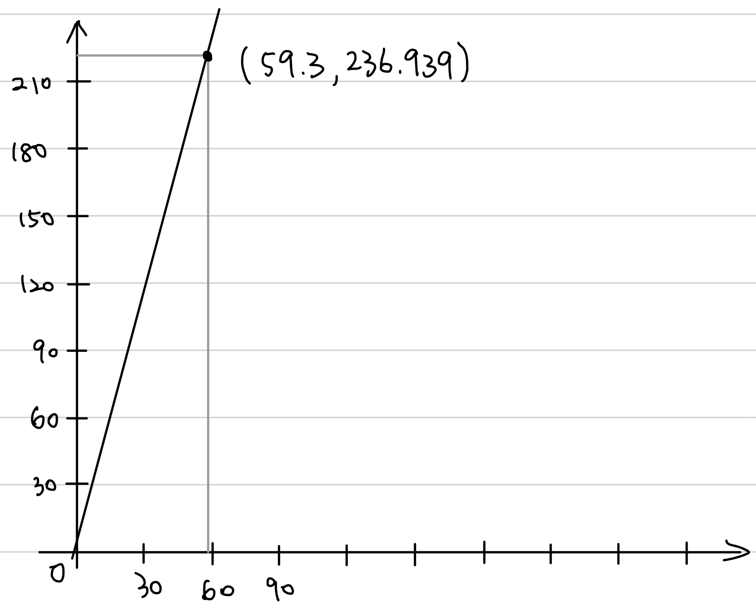
3.18 A life insurance company examines the relationship between the amount of life insurance held by a household and household income. Let *INCOME* be household income (thousands of dollars) and *INSURANCE* the amount of life insurance held (thousands of dollars). Using a random sample of $N = 20$ households, the least squares estimated relationship is

$$\widehat{INSURANCE} = \underset{(7.383)}{6.855} + \underset{(0.112)}{3.880}INCOME \quad \text{(se)}$$

a. Draw a sketch of the fitted relationship identifying the estimated slope and intercept. The sample mean of *INCOME* = 59.3. What is the sample mean of the amount of insurance held? Locate the point of the means in your sketch.

b. How much do we estimate that the average amount of insurance held changes with each additional \$1000 of household income? Provide both a point estimate and a 95% interval estimate. Explain the interval estimate to a group of stockholders in the insurance company.

a.



The sample mean of the amount of insurance is
$6.855 + 3.880 \cdot 59.3 = \underline{236.939}$.

b. Point estimate of b_2 is 3.880. So the point estimate of the average amount of insurance held changes with each additional \$1000 of household income is \$3880.
Next, we find the 95% confidence interval of b_2.

$$0.95 = P\left(t_{18;0.025} < \frac{b_2 - 3.88}{se(b_2)} < t_{18;0.975}\right)$$

$$= P\left(0.112 \cdot (-2.101) + 3.88 < b_2 < 0.112 \cdot 2.101 + 3.88\right)$$

$$= P(3.644 < b_2 < 4.115)$$

So the 95% C.I. of b_2 is $[3.644, 4.115]$, and 95% interval estimate of the average amount of insurance held changes with each additional \$1000 of household income is $[3644, 4115]$

(in \$1000 unit.)

c. Construct a 99% interval estimate of the expected amount of insurance held by a household with \$100,000 income. The estimated covariance between the intercept and slope coefficient is −0.746.

d. One member of the management board claims that for every \$1000 increase in income the average amount of life insurance held will increase by \$5000. Let the algebraic model be $INSURANCE = \beta_1 + \beta_2 INCOME + e$. Test the hypothesis that the statement is true against the alternative that it is not true. State the conjecture in terms of a null and alternative hypothesis about the model parameters. Use the 5% level of significance. Do the data support the claim or not? Clearly, indicate the test statistic used and the rejection region.

e. Test the hypothesis that as income increases the amount of life insurance held increases by the same amount. That is, test the null hypothesis that the slope is one. Use as the alternative that the slope is larger than one. State the null and alternative hypotheses in terms of the model parameters. Carry out the test at the 1% level of significance. Clearly indicate the test statistic used, and the rejection region. What is your conclusion?

c. We first compute $\widehat{var}(b_1 + 100b_2)$

$$\widehat{var}(b_1 + 100b_2) = \widehat{var}(b_1) + 100^2\,\widehat{var}(b_2) + 2\cdot 100\cdot\widehat{cov}(b_1, b_2)$$

$$= 7.383^2 + 10000\cdot 0.112^2 + 2\cdot 100\cdot(-0.746)$$

$$= 30.7487$$

So $se(b_1 + 100b_2) = \sqrt{\widehat{var}(b_1 + 100b_2)} = 5.545$

Thus, the 99% interval estimate of the expected amount of insurance held by a household with \$100,000 income is

$$6.855 + 3.88\cdot 100 \pm t_{18;0.995}\cdot se(b_1 + 100b_2)$$

$$= 394.855 \pm 2.878\cdot 5.545$$

$= 394.855 \pm 15.959$ in \$1000 unit or 394855 ± 15959 in \$1 unit.

d. ① $H_0: \beta_2 = \frac{5000}{1000} = 5$, $H_a: \beta_2 \neq 5$

② $\alpha = 0.05$

③ Test statistic: $t = \dfrac{b_2 - \beta_2}{se(b_2)} \sim t_{N-2}$

④ Realized statistic: $t^* = \dfrac{3.88 - 5}{0.112} = -10$

⑤ Rejection Region: $\{t : t < t_{18;0.025} \text{ or } t > t_{18;0.975}\}$
$= \{t : t < -2.101 \text{ or } t > 2.101\}$

⑥ Since $t^* = -10 < -2.101$, it falls in the rejection region ⇒ We reject H_0.
So the data doesn't support the claim.

e. ① $H_0: \beta_2 = 1$, $H_a: \beta_2 > 1$

② $\alpha = 0.01$

③ Test statistic: $t = \frac{b_2 - \beta_2}{se(b_2)} \sim t_{N-2}$

④ Realized statistic: $t^* = \frac{3.88 - 1}{0.112} = 25.71$

⑤ Rejection Region: $\{t : t > t_{18; 0.99}\}$
$= \{t : t > 2.552\}$

⑥ Since $t^* = 25.71 > 2.552$, it falls in the rejection region ⇒ We reject H_0.
So the data doesn't support the claim.

3.23 The data file *collegetown* contains data on 500 single-family houses sold in Baton Rouge, Louisiana, during 2009–2013. The data include sale price in \$1000 units, *PRICE*, and total interior area in hundreds of square feet, *SQFT*.

a. Using the quadratic regression model, $PRICE = \alpha_1 + \alpha_2 SOFT^2 + e$, test the hypothesis that the marginal effect on expected house price of increasing the size of a 2000 square foot house by 100 square feet is less than or equal to \$13000 against the alternative that the marginal effect will be greater than \$13000. Use the 5% level of significance. Clearly state the test statistic used, the rejection region, and the test p-value. What do you conclude?

b. Using the quadratic regression model in part (a), test the hypothesis that the marginal effect on expected house price of increasing the size of a 4000 square foot house by 100 square feet is less than or equal to \$13000 against the alternative that the marginal effect will be greater than \$13000. Use the 5% level of significance. Clearly state the test statistic used, the rejection region, and the test p-value. What do you conclude?

a. $\frac{dPRICE}{dSQFT} = 2\alpha_2 SQFT$. So

① $H_0: \alpha_2 = \frac{13}{2 \cdot 20} = 0.325$, $H_a > 0.325$

② $\alpha = 0.05$

③ Test Statistic : $t = \frac{a_2 - \alpha_2}{se(a_2)} \sim t_{N-2}$

④ Realized Statistic : $t^* = \frac{0.184519 - 0.325}{0.005255801} = -26.72875$

⑤ Rejection Region : $\{t : t > t_{df;0.05}\} = \{t : t > 1.647919\}$

p-value : $p = 1$

⑥ Since t^* doesn't fall in the rejection region, we can't reject the null hypothesis.

```
> # Problem a
> library(POE5Rdata)
> data("collegetown")
> ?collegetown
>
> # PRICE  = alpha1+alpha2*SQFT^2 + e
> mod1 <- lm(price~I(sqft^2),data=collegetown)
> smod1 <- summary(mod1)
> smod1

Call:
lm(formula = price ~ I(sqft^2), data = collegetown)

Residuals:
    Min      1Q  Median      3Q     Max
-383.67  -48.39   -7.50   38.75  469.70

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) 93.565854   6.072226   15.41   <2e-16 ***
I(sqft^2)    0.184519   0.005256   35.11   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 92.08 on 498 degrees of freedom
Multiple R-squared:  0.7122,  Adjusted R-squared:  0.7117
F-statistic:  1233 on 1 and 498 DF,  p-value: < 2.2e-16

> a1 <- coef(mod1)[1]
> a2 <- coef(mod1)[2]
> a2
I(sqft^2)
 0.184519
> sea1 <- coef(smod1)[1,2]
> sea2 <- coef(smod1)[2,2]
> sea2
[1] 0.005255801
> tstar <- (a2-0.325)/sea2
> tstar
I(sqft^2)
-26.72875
> alpha <- 0.05
> df = df.residual(mod1)
> tc <- qt(1-alpha,df)
> tc
[1] 1.647919
> p <- 1-pt(tstar,df)
> p
I(sqft^2)
        1
```

b. ① $H_0: \alpha_2 = \frac{13}{2 \cdot 40} = 0.1625$, $H_a > 0.1625$

② $\alpha = 0.05$

③ Test Statistic : $t = \frac{a_2 - \alpha_2}{se(a_2)} \sim t_{N-2}$

④ Realized Statistic : $t^* = \frac{0.184519 - 0.1625}{0.005255801} = 4.189467$

⑤ Rejection Region : $\{t : t > t_{df;0.05}\} = \{t : t > 1.647919\}$

p-value : $p = 1.65395 \times 10^{-5}$

⑥ Since t^* falls in the rejection region or $p < 0.05$, we reject the null hypothesis that the mariginal effect on expected house price of increasing the size of a 2000 square feet house by 100 square feet is less than or equal to \$13000.

```
> # Problem b
> tstar <- (a2-0.1625)/sea2
> tstar
I(sqft^2)
 4.189467
> alpha <- 0.05
> df = df.residual(mod1)
> tc <- qt(1-alpha,df)
> tc
[1] 1.647919
> p <- 1-pt(tstar,df)
> p
 I(sqft^2)
1.65395e-05
```

c. Using the quadratic regression model in part (a), estimate the expected price $E(PRICE|SQFT) = \alpha_1 + \alpha_2 SQFT^2$ for a house of 2000 square feet. Construct a 95% interval estimate of the expected price. Describe your interval estimate to a general audience.

d. Locate houses in the sample with 2000 square feet of living area. Calculate the sample mean (average) of their selling prices. Is the sample average of the selling price for houses with $SQFT = 20$ compatible with the result in part (c)? Explain.

c. $E(PRICE \mid SQFT = 2000/100)$
$= 167.3735$

So the estimated price for a house of 2000 square feet is
$167.3735 \times \$1000 = \$167373.5$

The 95% interval estimate of the expected price is
[155.5941, 179.1528] (in \$1000 unit) or
[155594.1, 179152.8] (in \$1 unit)

It means that in repeated sampling, about 95% intervals constructed this way will contain the true value of the price.

```
> # problem c
> x = 2000/100 # SQFT is in 100 square feet
> a1
(Intercept)
   93.56585
> a2
I(sqft^2)
 0.184519
> vara1 <- vcov(mod1)[1,1]
> vara2 <- vcov(mod1)[2,2]
> cova1a2 <- vcov(mod1)[1,2]
> L <- a1+a2*x^2
> L
(Intercept)
   167.3735
> varL = vara1 + x^2*vara2+ 2*x*cova1a2
> seL <- sqrt(varL)
> alpha <- 0.05
> tcr <- qt(1-alpha/2,df)
> lowbL <- L - tcr*seL
> upbL <- L + tcr*seL
> lowbL
(Intercept)
   155.5941
> upbL
(Intercept)
   179.1528
```

d.

```
> # Problem d
> collegetown$price[which(collegetown$sqft==20)]
[1] 138 169 183
> p20 <- mean(collegetown$price[which(collegetown$sqft==20)])
> p20
[1] 163.3333
```

The sample mean of the house with SQFT = 20 is 163.3333
Since 163.3333 is in the interval [141.5496, 163.7206], it is compatible with the result in part C.
This is not a surprise result, because in repeated sampling, about 95% intervals constructed this way will contain the true value of the price.

3.31 Data on weekly sales of a major brand of canned tuna by a supermarket chain in a large midwestern U.S. city during a mid-1990s calendar year are contained in the data file *tuna*. There are 52 observations for each of the variables. The variable $SAL1$ = unit sales of brand no. 1 canned tuna, and $APR1$ = price per can of brand no. 1 tuna (in dollars).

a. Calculate the summary statistics for $SAL1$ and $APR1$. What are the sample means, minimum and maximum values, and their standard deviations. Plot each of these variables versus $WEEK$. How much variation in sales and price is there from week to week?

b. Plot the variable $SAL1$ (y-axis) against $APR1$ (x-axis). Is there a positive or inverse relationship? Is that what you expected, or not? Why?

c. Create the variable $PRICE1 = 100APR1$. Estimate the linear regression $SAL1 = \beta_1 + \beta_2 PRICE1 + e$. What is the point estimate for the effect of a one cent increase in the price of brand no. 1 on the sales of brand no. 1? What is a 95% interval estimate for the effect of a one cent increase in the price of brand no. 1 on the sales of brand no. 1?

d. Construct a 90% interval estimate for the expected number of cans sold in a week when the price per can is 70 cents.

a.

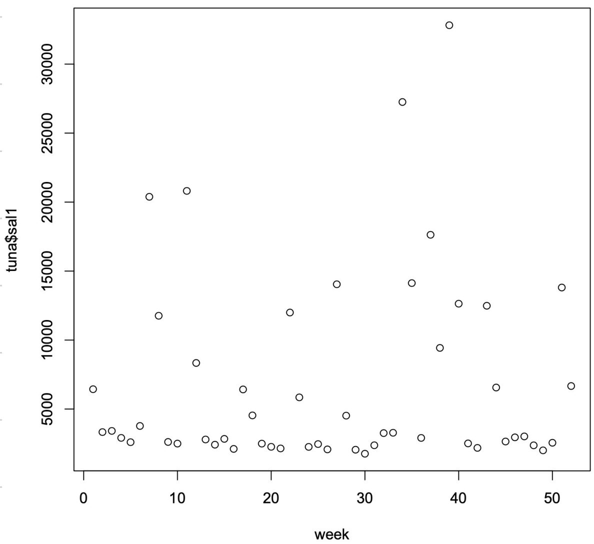
	Sample mean	Min.	Max.	Standard deviation
SAL1	6719	1762	32820	6915.083
APR1	0.7825	0.61	0.92	0.09803711

```
> # Problem a
> library(POE5Rdata)
> data("tuna")
> ?tuna
> summary(tuna$sal1)
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
   1762    2485    3136    6719    8612   32820
> sd(tuna$sal1)
[1] 6915.083
> summary(tuna$apr1)
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
 0.6100  0.6900  0.8100  0.7825  0.8625  0.9200
> sd(tuna$apr1)
[1] 0.09803711
```

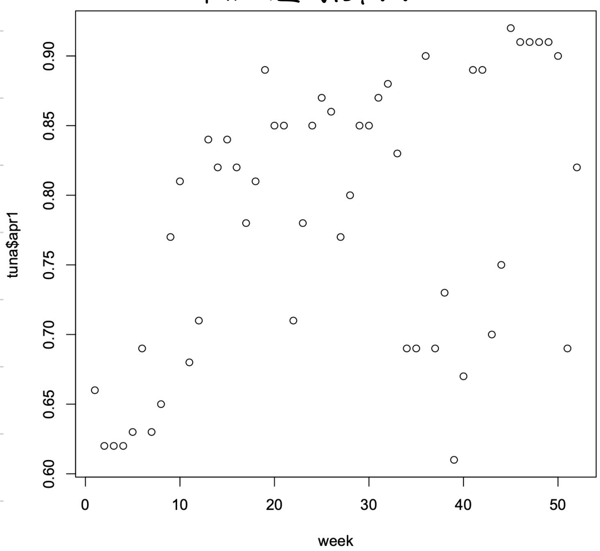
CV of SAL1: $\frac{6915.083}{6719} = 1.0292$

CV of APR 1: $\frac{0.09803711}{0.7825} = 0.1253$

SAL1 v.s. WEEK



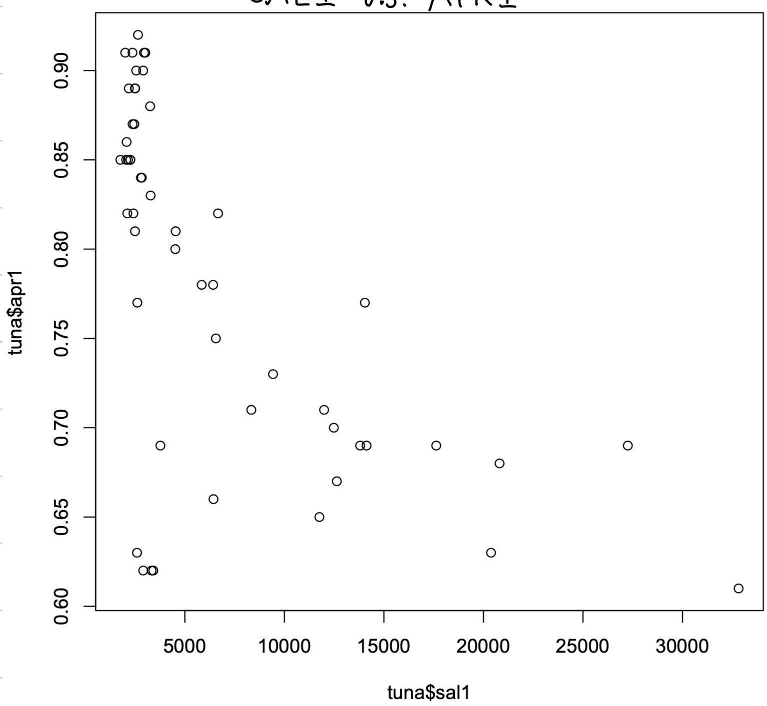
APR1 v.s. WEEK



Sales (SAL1) exhibit substantial week-to-week variation, with most observations clustered at low levels but occasional large spikes.
In contract, price (APR1) shows relatively small and smooth fluctuations over time, with no extreme values, indicating lower variability compare to sales.

b.

SAL1 v.s. APR1



There is an inverse relationship, which makes sense because of the law of demand.

C. Point estimate for the effect of a one cent increase in the price of brand no.1 on the sales of brand no.1 is $b_2 = -434.4473$

A 95% interval estimate of b_2 is $[-592.2897, -276.6049]$.

```
> # Problem c
> price1 = 100*tuna$apr1
> mod1 <- lm(tuna$sal1~price1)
> smod1 <- summary(mod1)
> smod1

Call:
lm(formula = tuna$sal1 ~ price1)

Residuals:
     Min       1Q   Median       3Q      Max 
-10869.5  -1588.3   -499.3   1626.2  18607.1 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept) 40714.22    6196.42   6.571 2.82e-08 ***
price1       -434.45      78.58  -5.528 1.17e-06 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 5502 on 50 degrees of freedom
Multiple R-squared:  0.3794,	Adjusted R-squared:  0.367 
F-statistic: 30.56 on 1 and 50 DF,  p-value: 1.172e-06
```

```
> b2 <- coef(mod1)[2]
> b2
   price1 
-434.4473 
> seb2 <- coef(smod1)[2,2]
> alpha <- 0.05
> df = df.residual(mod1)
> tcr <- qt(1-alpha/2,df)
> lowbb2 <- b2 - tcr*seb2
> upbb2 <- b2 + tcr*seb2
> lowbb2
   price1 
-592.2897 
> upbb2
   price1 
-276.6049 
```

d. When the price is 70 cents, the 90% interval estimate for the expected number of cans sold in a week is
[8624.934, 11980.87]

```
> # problem d
> x = 70
> b1
(Intercept)
   40714.22
> b2
   price1
-434.4473
> varb1 <- vcov(mod1)[1,1]
> varb2 <- vcov(mod1)[2,2]
> covb1b2 <- vcov(mod1)[1,2]
> L <- b1+b2*x
> L
(Intercept)
    10302.9
> varL = varb1 + x^2*varb2+ 2*x*covb1b2
> seL <- sqrt(varL)
> alpha <- 0.1
> tcr <- qt(1-alpha/2,df)
> lowbL <- L - tcr*seL
> upbL <- L + tcr*seL
> lowbL
(Intercept)
   8624.934
> upbL
(Intercept)
   11980.87
```

e. Construct a 95% interval estimate of the elasticity of sales of brand no. 1 with respect to the price of brand no. 1 "at the means." Treat the sample means as constants and not random variables. Do you find the sales are fairly elastic, or fairly inelastic, with respect to price? Does this make economic sense? Why?

f. Test the hypothesis that elasticity of sales of brand no. 1 with respect to the price of brand no. 1 from part (e) is minus three against the alternative that the elasticity is not minus three. Use the 10% level of significance. Clearly, state the null and alternative hypotheses in terms of the model parameters, give the rejection region, and the p-value for the test. What is your conclusion?

e. $$\text{Elasticity} = \frac{\Delta y / y}{\Delta x / x}$$

$$= \beta_2 \cdot \frac{\text{PRICE1}}{\text{SAL1}}$$

A 95% interval estimate of the elasticity is [-6.898149, -3.221501]. So it is fairly elastic with respect to price. This makes economic sense because of the law of demand.

```
> # Problem e
> pricemean <- mean(price1)
> pricemean
[1] 78.25
> salmean <- mean(tuna$sal1)
> salmean
[1] 6718.712
> elas <- b2 * pricemean/salmean
> elas
   price1
-5.059825
> alpha <- 0.05
> df = df.residual(mod1)
> tc <- qt(1-alpha/2,df)
> seelas = pricemean/salmean * seb2
> lowelas <- elas - tc  * seelas
> upelas <- elas + tc  * seelas
> lowelas
   price1
-6.898149
> upelas
   price1
-3.221501
```

f. ① $H_0: \varepsilon = -3$ $H_a: \varepsilon \neq -3$

② $\alpha = 0.1$

③ Test statistic:

$$\frac{\varepsilon - (-3)}{se(\varepsilon)} \sim t_{N-2}$$

④ Realized statistic:

$$t^* = \frac{-5.059825 - (-3)}{0.9152451} = -2.250572$$

⑤ Rejection Region:

$$\{t : t < t_{df;0.05} \text{ or } t > t_{df;0.95}\}$$

$$= \{t : t < -1.675905 \text{ or } t > 1.675905\}$$

p-value: $p = 0.02884204$

⑥ Since -2.250572 falls in the rejection region, we conclude that we reject the null hypothesis $\varepsilon = -3$.

```
> # Problem f
> alpha <- 0.1
> tstar = (elas-(-3))/seelas
> elas
   price1
-5.059825
> seelas
[1] 0.9152451
> tstar
   price1
-2.250572
> tc <- qt(1-alpha/2,df)
> tc
[1] 1.675905
> lowelas <- elas - tc  * seelas
> upelas <- elas + tc  * seelas
> lowelas
   price1
-6.593689
> upelas
   price1
-3.525961
> p <- 2*pt(tstar,df)
> p
    price1
0.02884204
```